



UNIVERSITY
OF COLOGNE

WORKSHOP 07

Advanced Analytics and Applications

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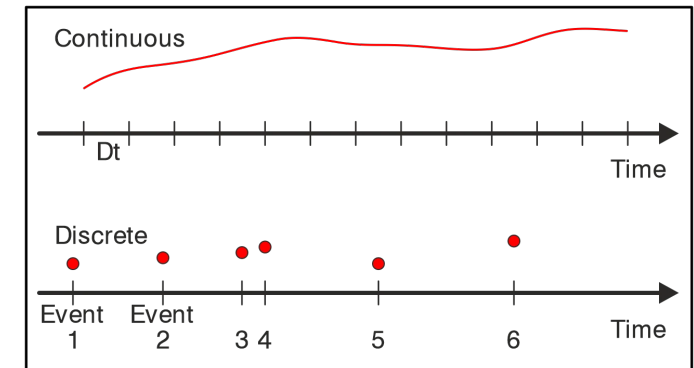
Coding Tasks

Systems, Models, Simulations

- A **system** is anything that consists of multiple, interacting elements. Elements can be passive (resources, objects, etc) or active (humans, agents, etc) and the output of the system is a result of the interaction between elements within.
 - Application: Software components doing what they were programmed to do.
 - Information System: IT and humans in an organization, doing processes together.
 - Society: Humans and regulatory frameworks (laws, incentives, ...) interacting.
- A **model** is any abstraction (no matter how high or low level) of a system. Usually, a model has a focus.
- For example, elevators in a building can be modelled with different foci:
 - a) Focus on flow: How long does it take to transport humans and things, how many meters do elevators do a day, how intelligent is the control system, can emergencies overrule?
 - b) Focus on energy: How much electricity do the elevators draw depending on load? Can the inertia be used to create electrical energy? Does the control system consider power limits?

Systems, Models, Simulation

- A **simulation** is the imitation of the system using the model. The simulation merely “runs” the modelled abstraction using whatever processing technique is used (which, again, depends on the model)
- It executes every modelled interaction between entities and progresses through time steps using either time progression or event progression.
 - *Time progression* means the simulation clock runs forward for t seconds/minutes/..., we progress any entities and calculate any metrics from x_t to x_{t+1} , and we observe the system state at this point
 - *Event progression* means we look at what start/stop/progress conditions are currently met and advance the queue of events to the next checkpoint and observe the system state at this point.
Time is usually also modelled here, but it is not an active component, rather a bookkeeper. This is called “next-event time progression”.



Agent-Based Modelling (ABM)

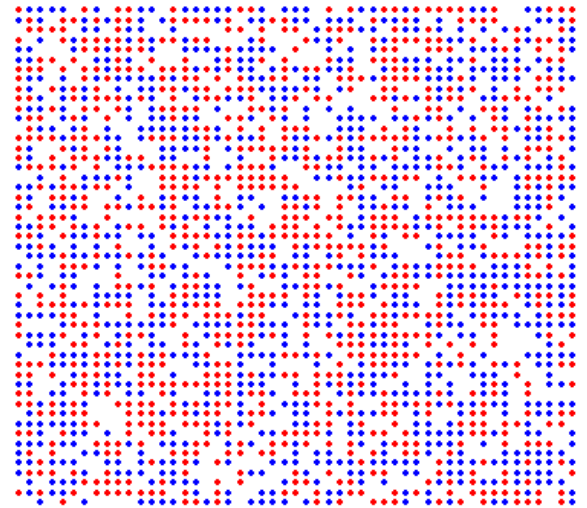
- Different from top-down models that assign stochastic patterns to parts of the simulation, agent-based models define **agents with own behavior** and their interaction to simulate bottom-up.
- It helps to capture **complex interaction** between hundreds or thousands of small/naïve agents. From them, **patterns can emerge** that we wouldn't be able to impose as rules to the simulation model.
- The emergent patterns are not properties of any single agent but the **system** of them. It's **not possible to simply average** behaviors of single agents to a system behavior.

Agent-Based Modelling: Example

One of the most famous examples is that of “Schelling’s World”, which shows how macro-level **segregation** emerges from micro-level preferences that are not explicitly segregation:

1. Place agents on a map and randomly assign one of two groups.
2. Set a threshold of moving to a different neighborhood that could very well be realistic: Most people like diversity as long as they are not in a strict minority themselves: Agent is happy if the fraction of same group in their neighborhood is above or equal some τ
3. Each round, agents look around and stay where they are if the fraction of same-group neighbors is $\geq \tau$
Otherwise, they will relocate to another, random position.

Step 1



GIF: <https://2.bp.blogspot.com/-oILtOwdFc/T50Bt4Jw0hI/AAAAAAAAADk/IC2cKyC42gA/s1600/animation.gif>

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Question 1: Modelling Systems

Question: Consider the following examples and state which type of model they would necessitate.

A) Wind tunnel for testing aerodynamics of cars.



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Answer: This is a dynamic (time plays a role), deterministic, and continuous system and thus best modelled by an **analytical** model (set of equations)

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B) A system of multiple elevators in a hotel that coordinate among themselves upon being periodically called from the lobby / ground floor.

Answer: This is a typical example of a **discrete event simulation** (dynamic, with discrete time steps) because it is stochastic (we only have distributional information about expected “button presses” in the lobby and can thus model it using an event generation process, Poisson e.g.)

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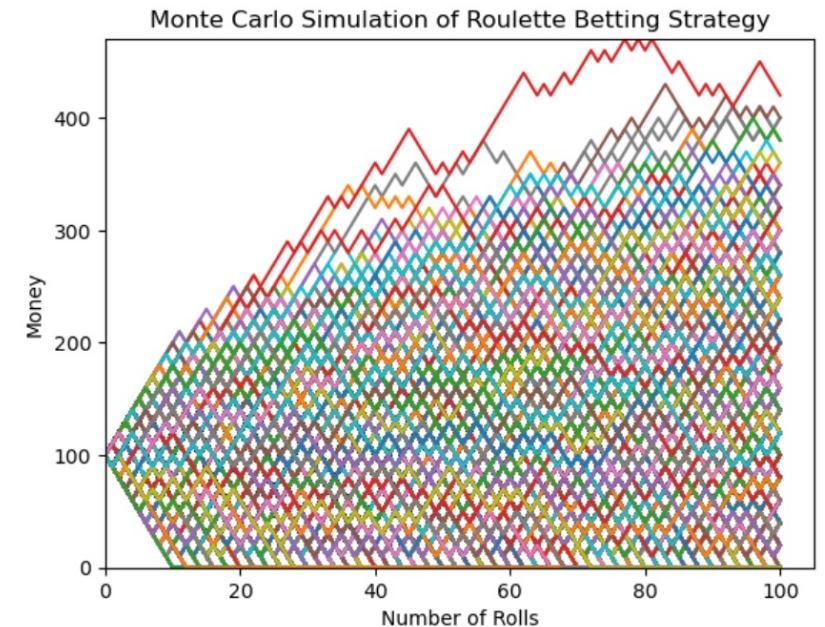
Answer: The system is **stochastic** (win probabilities for colors/numbers) but it is **static** with regard to time: It doesn't matter if I wait ten seconds or two hours between two rounds!
This is a typical use case for **Monte Carlo** simulation, even though the name is just loosely associated with casino simulation.

Question 2: Reproducibility

Question: You just built a nice model of playing roulette X times in a row. A colleague simulates a couple of runs for the same X and tells you the model doesn't perform as expected, the results are different.

Explain to your colleague what he got wrong in understanding how simulations work.

Answer: When we simulate a stochastic model, getting different results for the same inputs, is not a bug but a feature of that 😊 Stochastic models draw from a parametrized distribution every time. This means that (by the law of large numbers), after a lot of runs, we should arrive at an **expected result** but that doesn't tell us anything about a single run.



Question 3: Experiments

Question: In the following example, what did the researchers not consider? What should they have done?

“New York City introduced a congestion price (a toll) in the Central Business District in January. Researchers from a local transport institute considered that to be the perfect natural experiment and began collecting data on how many vehicles entered the Central Business District and compared that with data from before and found that the reduce was dramatic.”

Answer: The researchers compare the data with data from before, which is only half of a natural experiment. They forgot that change over time can be driven by any other factors and not exclusively attributed to that particular change. They should have done the exact same comparison with previous data for another spatial unit (which is comparable in terms of factors that influence it over time) to compare the change (“difference in differences” method).

For an example, see Abadie, A., Diamond, A., Hainmueller, J., 2010. Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California’s Tobacco Control Program. Journal of the American Statistical Association 105, 493–505. <https://doi.org/10.1198/jasa.2009.ap08746>

Question 4: ABM

Question: Give two examples for simulation where you **don't** want to use agent-based modelling and explain why.

Answer:

- 1) Well-understood dynamics, e.g. fluid dynamics. You **can** model it directly through equations and you can even solve and analyze it analytically. ABM would introduce unnecessary and even bad overhead and hinder the ability to analyze the continuum.
- 2) Models that can only be analyzed in an aggregate manner because of size: Census forecasting, economic growth etc of a country with a millions of inhabitants. Arriving at the aggregate, system-level metrics would mean you need to be able to configure all the inhabitants with heterogeneous behaviors.

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SimPy: Discrete Event Simulation Framework



- In this class, we will be using SimPy as the framework for building simulations.
- You can access its documentation here: <https://simpy.readthedocs.io/en/latest/>
- The underlying model, that you want to simulate, can be any arbitrary Python code, as long as it you can boil it down to **processes, events, and resources**.
- This, on the other hand, means that you have to build whatever you want to simulate yourself. SimPy is merely a framework (like PyTorch for Deep Learning).

Contact



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